Generation and editing of faces using Stable Diffusion with Criminal Suspect Matching

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***Abstract*—Traditional criminal identification is done via verbal communication of features of the criminal and subsequent sketching. Sketches drawn are generally limited by the artistic talent of the artist and prone to human errors. The solution proposed to the problem is using image generation to provide a realistic and high quality image rather than a black and white sketch. However there is a possibility that the generated image does not match the image envisioned by the witness and might not look exactly alike to a real person .This paper proposes a novel approach combining components to generate realistic faces through a textual description and editing the generated face further through textual description using stable diffusion. The final image generated is matched to a dataset to provide a real image that is similar to the generated image.The current state-of-the-art method for image synthesis through text involves using Generative adversarial network(GAN). The proposed method uses a diffusion based model for generating and editing of the image through text prompt that provides more fine grained control to image synthesis and high quality images for matching with a dataset.**

***Index Terms*—Stable Diffusion, Face Generation,Face Editing, Matching.**

I. Introduction

In recent years, the crime rates have increased in many parts of the world. In each of these cases, images of criminals have played an important role in helping to identify the criminal and bring justice and hence protect the public. Eyewitness descriptions are often the primary source of information to identify a suspect in the occurrence of a crime. These descriptions are subjective and often don't provide a clear idea on the suspect's visual representation. Face generation making use of the artificial neural networks offers the potential to improve the accuracy and reliability of suspect identification. In most cases, the face generated may not be the face the eyewitness envisioned or may not belong to a real person. The paper proposes the generation of photo-realistic images of faces based on the description of the suspect's facial features and editing of the image with further text. Image generation with GANs have training instabilities and mode collapse. This paper explores and makes use of stable diffusion, a relatively new algorithm for image generation and editing, giving more control to the output. The facial image is finally matched against a database of criminal suspects.

II. Related Work

Generation of images from textual input by use of artificial neural networks has been a widely explored field since the mid 2010s. GANs were introduced in 2014 and this played an influential role in advancing text-to-image generation. However, this had its own set of challenges as high-resolution and realistic images with fine details and the inability to handle nuanced textual descriptions. The training instabilities encountered and the mode collapse brought to the development of a new algorithm, the stable diffusion.

The stable diffusion algorithm applied a gaussian noise to the image and trained the model to reduce this noise thus giving more control to the output generated. Different types of diffusion models now exist in the field and are continued to be worked upon to achieve better results.

Wavelet Diffusion Models [4], based on the idea of iteratively applying a diffusion process to a noise vector, transforms into an image at each iteration. They are fast and scalable image generators which involve adding noise to the image and then filtering it using a wavelet transform. This process is repeated for a number of iterations until the image is fully generated. One of the advantages is their speed and scalability. Unlike other generative models such as GANs, WDMs do not require a complex training process or a large amount of computational resources. This makes them suitable for generating images in real-time applications or on low-power devices. While WDMs are relatively fast to train compared to some other generative models, they still require careful tuning of several hyperparameters, including the number of iterations, the diffusion process parameters,

and the wavelet transform parameters.

High resolution image synthesis with latent diffusion methods [2] utilizes deep learning training within the latent space of highly-capable pre-trained autoencoders. This makes use of cross-attention layers which are flexible generators. The architecture links transformers to the DM's U-Net backbone, allowing for diverse token-based conditioning techniques. This approach notably decreases computational demands in comparison to pixel-based DMs, while also minimizing its carbon footprint making it less expensive in time and memory. However, the sequential sampling process remains slower in comparison to that of GANs and the image reconstruction capability posed a limitation for tasks that demand precise accuracy in pixel space.The model sometimes misrepresented the data as well.

Blended diffusion for text-driven editing [16] of natural images works by first encoding the image and text into two separate latent spaces using pretrained models. These latent representations are then used to initialize two diffusion processes, one for the image and one for the text. The diffusion processes iteratively update the latent representations, gradually blending the image and text information. The method is easy to use and requires minimal user input and produces high-quality and realistic results, even when the text descriptions are ambiguous or incomplete. The limitation of this method is that it relies on diffusion processes that blend the image and text information, which can make it difficult to have fine-grained control over the editing process. While the user can provide a text description of the desired change, the method may not always produce the exact result that the user had in mind.

The Stable Artist: Steering Semantics in Diffusion Latent Space [1] proposes a semantic guidance (SEGA) model which enables nuanced image editing, alterations to composition, and adjustments to style. It facilitates the exploration of latent spaces to gain a better understanding of the acquired concept representations and captures statistical properties of text data and creates a latent space that can be manipulated. Classifier-free guidance is used along with editing concepts to combine text conditioning. It produces high quality output while maintaining the coherence and fluency of the generated text. By utilizing advanced operations, it grants precise control over the process of image generation without making use of masks and requires no fine-tuning. Unlike token-based attention, SEGA does not depend solely on natural language and enables conditioning beyond it. However, the model is complex and has limited generalisability as it is highly dependent on the dataset. It relies on noisy, large scale datasets or web crawled data.

Photorealistic text-to-image diffusion models with deep language understanding [11] proposes a novel text encoder that can effectively capture the semantics of the input text to incorporate deep language understanding into the image generation process. This text encoder is based on a combination of transformers and convolutional neural networks and is trained to map the input text to a high-dimensional feature vector. The novel text encoder can effectively capture the semantics of the input text. The method is scalable and can generate images for large datasets. However, this method can be computationally expensive and time-consuming as it relies heavily on textual descriptions to generate images. It generates images using a complex model that may be difficult to interpret.

Survey on diverse image inpainting using diffusion models [14] makes use of the diffusion process to generate a range of realistic and varied inpainted pictures for a particular image that has degraded. In this, the context of the surrounding area is used to repeatedly update the missing pixels. Data may be used to determine the parameters that govern the diffusion process. Diffusion models have the benefit of handling vast missing regions and yet generating results that look visually convincing.This makes it a helpful tool for erasing things from photos or fixing damaged photographs. The computational efficiency of SDM is an additional benefit. It has been demonstrated to work faster and more accurately than other cutting-edge inpainting algorithms. Additionally adaptable, SDM can work with a variety of picture textures, such as grass, fur, and hair. It is difficult to train the deep neural network that produced the preliminary estimate of the missing pixels, though. Photos with many tiny, fine features or photos with extremely complex textures, including abstract images, might not perform as well for it. The original image's tiny features are difficult for the algorithm to capture. Due to the employment of two diffusion models—an iterative process—instead of GANs, which operate in a single step, there is a significant computing overhead.

GAN vs Diffusion: Instance-Aware Inpainting on Small Datasets [15] evaluates the effectiveness of two cutting-edge methods for instance-aware inpainting: instance-aware GAN (InstaGAN) and RePaint, a denoising diffusion probabilistic model using limited datasets. The results demonstrate that for small-scale instance-aware inpainting tasks, RePaint performs better than InstaGAN. Compared to InstaGAN's instance-aware GAN method, RePaint achieves superior results by successfully removing noise through the use of a diffusion process that represents the picture pixel values as a random walk. RePaint's versatility for inpainting jobs is further enhanced by the diffusion process, which allows it to handle a broad range of noise distributions.With a lower FID score and LPIPS score, the results quantitatively demonstrate that RePaint performs better than InstaGAN in small-scale instance-aware inpainting tasks. Despite the limited amount of the dataset, the DDPM model performed better in terms of picture quality and consistency than both InstaGAN and CycleGAN.The findings imply that diffusion-based models, such as DDPM, might be a useful path for instance-aware image-to-image translation,

particularly in the case of small or noisy/uncertain datasets.

III. Proposed Methodology

A. Data

The dataset used to train the generation model is Multi-Modal-CelebA-HQ (MM-CelebA-HQ). It contains 30000 high resolution face images with descriptive text of each image. This specific dataset was chosen because it contained accurate descriptions and high quality images of faces.

Face Matching: The face matching dataset comprises 100 criminal face images obtained from the web, and these images have not undergone preprocessing.

B. Face Generation

The Stable Diffusion model trained consists of three major components.The Variational Autoencoder(VAE) consists of a separate encoder and decoder. The encoder compresses the 512x512 pixel image into a smaller 64x64 model in latent space that's easier to manipulate and requires far less computation to work on. During the training process noise is

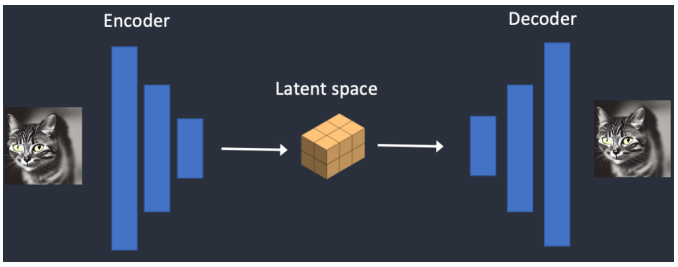


Fig. 1. Working of VAE

added gradually to the latent.These noisy latents are then taken as input to the U-Net. The text encoder is the component that transforms the input text prompt into embeddings. Contrastive Language-Image Pre-Training(CLIP) is used as the text encoder to convert the text prompt into text embeddings. The

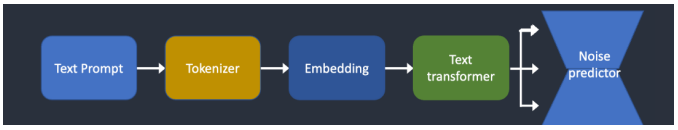


Fig. 2. Text prompt inputted into U-Net

embeddings are also taken as input to the U-Net . U-Net is a fully convolutional neural network(CNN) that is designed to learn from fewer training samples and mainly used for image segmentation.Image segmentation is the process of assigning a label to every pixel in an image such that pixels with the same label share certain characteristics The U-Net with the noisy latents and the text embeddings as input predicts a denoised image representation. This process is repeated for a certain number of sampling steps. To obtain a new image the predicted noise is subtracted from the latent image. Finally

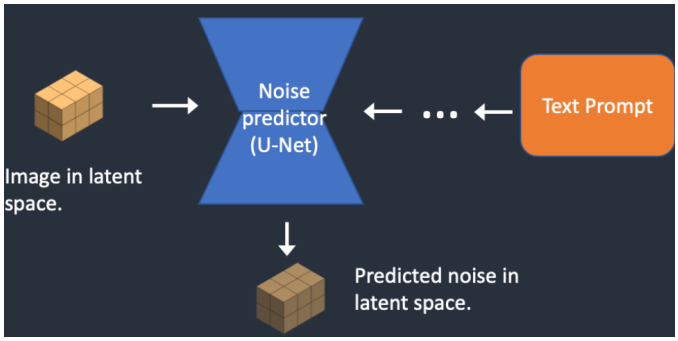


Fig. 3. Working of U-Net

the new latent image is fed into the decoder which transforms the latent representation back into a 512X512 pixel image. In

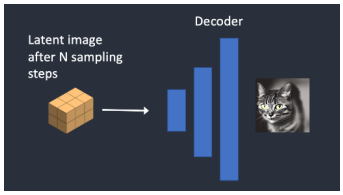


Fig. 4. New Latent image being decoded to create new image

the training process the model was trained by using a base model and further fine tuning the base model to output high quality faces. A base model was used to reduce computation and as a starting point of our model. Denoising diffusion probabilistic models(DDPM) scheduler was used to add noise to the image in the training process.A Learning Rate(LR) Scheduler is used for an adaptive learning rate to improve performance and reducing training time by reducing learning rate over time as the number of epochs increase. The images from MM-CelebA-HQ with captions were taken as input. The images is inputted into the VAE and the captions into CLIP to fine tune the model based on the image caption pairs of the dataset.Finally with text as the input it is converted into embeddings and the the image is generated by the VAE.

C. Face Manipulation

The manipulation model is trained to apply precise instructions to modify these facial photos while adhering to legal and ethical guidelines. Among the methodology's main points are:

1. Training of Stable Diffusion Models:

An enormous dataset of photos of suspected criminals is used to train the machine. Because of its inherent stability and controllability during the diffusion process, the Stable Diffusion model was selected.

2. Producing Lifelike Face Images:

Realistic facial images are created using the trained model. The created images function as a foundation for further alterations.

3. Manipulating Face Images:

Face feature manipulation is done according to predetermined guidelines. Standards of ethics and law are upheld throughout the manipulation process.

4. Verification of Manipulated Faces:

- A validation procedure is used to guarantee that the altered faces are accurate and realistic.

5. Hyperparameters:

The model's input consists of an image and a prompt. - There are three hyperparameters listed: - Number of Inference Steps: Indicates how many steps the diffusion model goes through to fine-tune the picture. The Image Guidance Scale regulates the original image's impact on the diffusion process. - Guidance Scale: Indicates how strong guidance is overall in the diffusion model.

6. Model Output Specifics:

The altered facial image is the main output of the model. The directions given have an impact on the manipulation process, and the model is educated to skillfully modify facial characteristics without sacrificing overall realism.

D. Face Matching

The reference image is loaded using the face recognition library. The facial features of the reference image are encoded to create a representation of the face. A directory containing generated images is specified. Each image in the directory is loaded for comparison. For each generated image, facial features are encoded using the same method as the reference image.

The proposed methodology computes a matching score by measuring the distance between the encoded features of the reference and generated images. The score is then normalized to a value between 0 and 1, where a higher value indicates a closer match. The proposed methodology keeps a track of the image with the highest matching score.

If a generated image has a higher score than the current highest score, it becomes the new best match. Once all images are processed, the script prints the highest matching score and the path to the best matching image. If a matching image is found, it is displaying using 'matplotlib' and the 'PIL'(python Imaging Library).

The matching score is a measure of how closely the facial features of the generated images resemble those of the reference image. A higher matching score suggests a better match. The threshold is set to 80 . If no faces are found in an image during encoding , it is mentioned , and that image is skipped.

The smaller distance in the face recognition library represents a better match, with a distance of 0 indicating an exact match. The matching score is computed as the inverse of this distance. The proposed methodology provides insights into the degree of similarity between the facial features of the reference and generated images, enabling the identification of the generated image that best matches the reference.

IV. Results and Discussion

A. Face Generation

The stable diffusion algorithm was trained on 1000 images the Multi-Modal-CelebA-HQ (MM-CelebA-HQ) and was run 100 epochs. Sample prompts were given during the training process which is shown in Fig 2 and 3. From these figures, it is seen how the features that were given as text prompt becomes more relevant in the images as the training proceeds.



Fig. 5. Sample of image generated during training process for prompt "long face man with french beard and wrinkled forehead".



Fig. 6. Sample of image generated during training process for prompt "red haired girl with high cheekbones and rosy cheeks".

The trained model was then used to generate the images. The figures 4 and 5 are some samples of the images generated by giving the textual description as input prompt.



Fig. 7. Sample of image generated by the trained model for prompt "long faced man with french beard and bushy eyebrows".

The CLIP score is obtained for the model and is shown in Fig 6.

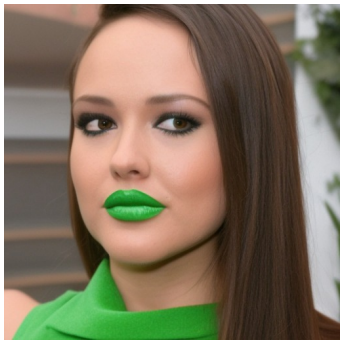


Fig. 8. Sample of image generated by the trained model for prompt ”woman with green lipstick and big forehead”.



Fig. 9. CLIP score obtained for the trained model and evaluated as images generated against prompts given.

B. Face Manipulation

The model is editing the features of the image based on the text prompt provided by the user. These changes are made to the image generated by the generation model.

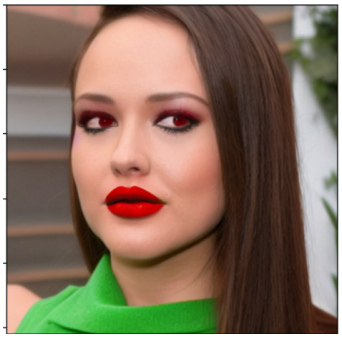


Fig. 10. Image generated by the manipulation model on editing the figure generated using prompt “make only her lips red”



Fig. 11. Image generated by the manipulation model on editing the figure generated using prompt “make her hair modern”



Fig. 12. Image generated by the manipulation model on editing the figure generated using prompt “make his hair white”



Fig. 13. Image generated by the manipulation model on editing the figure generated using prompt “reduce his beard hair”.

C. Face Matching

The manipulated image undergoes matching against an unprocessed dataset. The image exhibiting the highest matching score is presented, accompanied by the corresponding scores. A superior match is characterized by a higher score, with a threshold set at 80 or above to determine optimal matches.

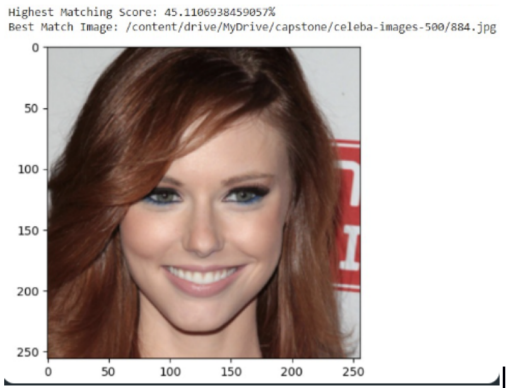


Fig. 14. Manipulated image matched against 8.2.1 Image of a woman in red lipstick along with the scores.

V. Conclusion and Future Work

The purpose of the proposed methodology is to generate faces and edit images through a text prompt using the novel



Fig. 15. Manipulated image matched against 8.2.3 image of a man in a white beard and white hair along with the scores .

method of stable diffusion. The generated image is then matched using resnet. An issue faced was that the image generated might not be a face if the text prompt is not related to the features of a face and for editing the parameters must be fine tuned depending on the image and the prompt had to be specific. The model trained initially was not adequate and higher computing specifications were required to train the model. The latent stable diffusion model outputs more realistic and high quality faces with the appropriate text prompt and was trained using a limited number of images. For future work inputting the prompt can be further controlled and the model can be further optimized to work with lesser computing specifications. In editing for future work the prompt can be made to include GPT-3 like models.

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